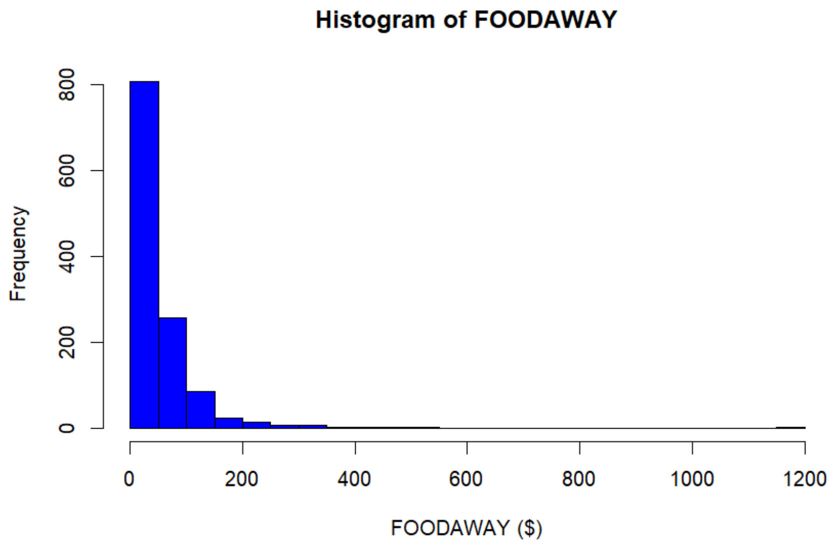
Consumer expenditure data from 2013 are contained in the file cex5_small. [Note: cex5 is a larger version with more observations and variables.] Data are on three-person households consisting of a husband and wife, plus one other member, with incomes between $1000 per month to $20,000 per month. FOODAWAY is past quarter's food away from home expenditure per month per person, in dollars, and INCOME is household monthly income during past year, in $100 units.

a. Construct a histogram of FOODAWAY and its summary statistics. What are the mean and median values? What are the 25th and 75th percentiles?



Min.	25th	median	75th	Max.	mean
0.00	12.04	32.55	67.5	1179.0	49.27

b. What are the mean and median values of FOODAWAY for households including a member with an advanced degree? With a college degree member? With no advanced or college degree member?

- with an advanced degree

	mean	median
advanced degree	73.15494	48.15

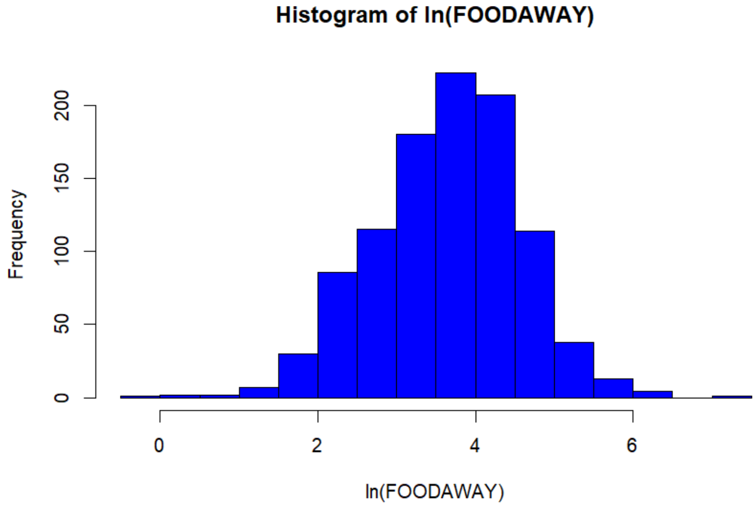
- with a college degree

	mean	median
college degree	48.59718	36.11

- With no advanced or college degree

	mean	median
no	39.01017	26.02

c. Construct a histogram of ln(FOODAWAY) and its summary statistics. Explain why FOODAWAY and ln(FOODAWAY) have different numbers of observations.



Min.	25th	median	75th	Max.	mean
-0.3011	3.0759	3.6865	4.2797	7.0724	3.6508

A histogram of ln(FOODAWAY) was created using only positive FOODAWAY values. The number of observations for ln(FOODAWAY) is smaller than FOODAWAY because households with zero FOODAWAY are excluded, as the natural logarithm of zero is undefined.

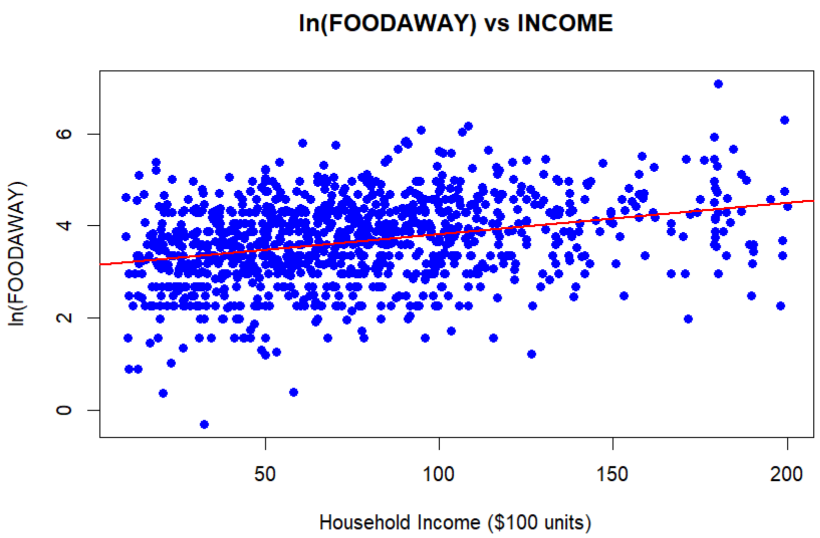
d. Estimate the linear regression ln(FOODAWAY)=β_1 +β_2INCOME + e. Interpret the estimated slope.

$\beta_1 \approx 3.1293$

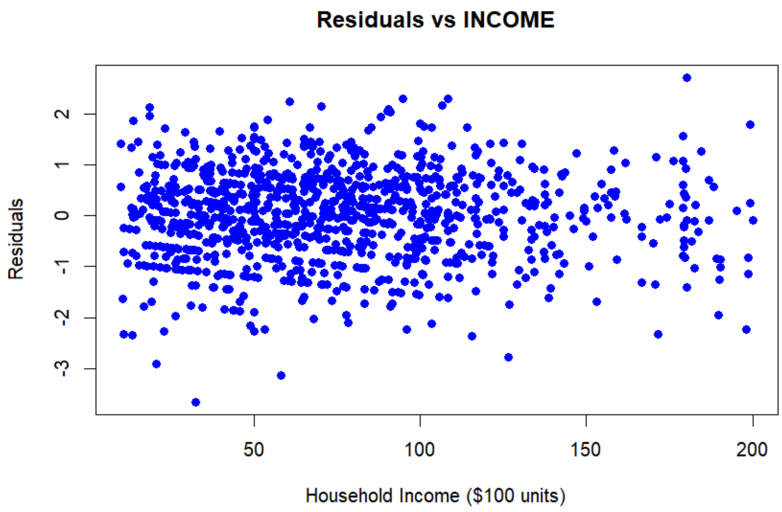
$\beta_2 \approx 0.0069017$

The estimated slope coefficient β2 ≈ 0.0069017 indicates that for every additional $100(one units) of household monthly income, the expected FOODAWAY expenditure per person increases by approximately 0.69%. Since the dependent variable is in logarithmic form, the slope represents the approximate percentage change in FOODAWAY for a one-unit increase in INCOME.

e. Plot ln(FOODAWAY) against INCOME, and include the fitted line from part (d).



f. Calculate the least squares residuals from the estimation in part (d). Plot them vs. INCOME. Do you find any unusual patterns, or do they seem completely random?



The residuals appear to be randomly scattered around zero, without any obvious pattern. This suggests that the linear model provides an adequate fit for the data.